

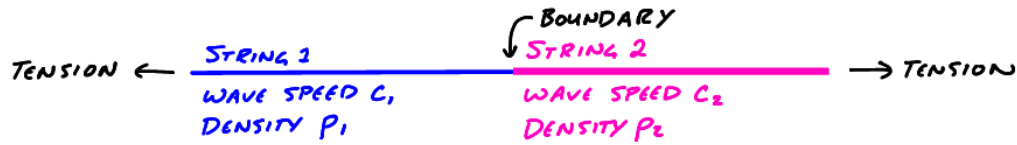
RESONANCE AKA. "FITTING WAVES INTO STRUCTURES"

- EXTREMELY CENTRAL TO THIS COURSE: IT'S HOW WE CREATE INSTRUMENTS!!
- CLAMPED IDEAL STRINGS (1D)
 - ↳ HARMONIC OSCILLATORS (IMPORTANT!)
- DRUMS (2D)
- PIPES + CHAMBERS (3D)

STRING RESONANCE BASICS

- AS USUAL, LET'S START WITH THE SIMPLEST WAVE SYSTEM: IDEAL STRINGS

- RECALL HOW WAVES INTERACT WITH A BOUNDARY:



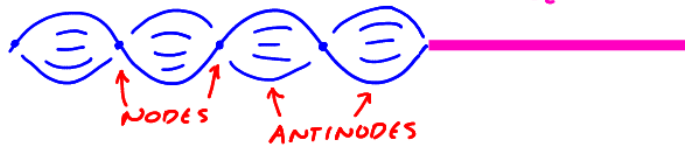
$$\Rightarrow H_R = \frac{c_2 - c_1}{c_2 + c_1} H_I$$

$$H_T = \frac{2c_2}{c_2 + c_1} H_I$$

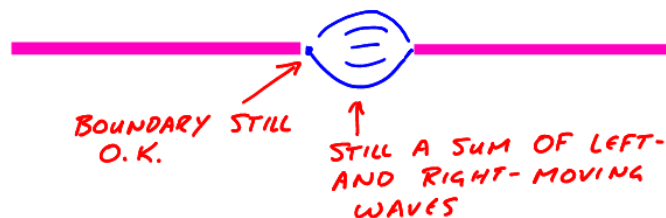
$\Rightarrow$ WHEN $\rho_2 \gg \rho_1$ ($c_2 \ll c_1$) THE BOUNDARY IS "CLAMPED"

- THIS FORCED THAT $H_R \rightarrow -H_I$ WITH RELATIVE PHASE = 0
SO THAT THE DISPLACEMENT AT THE BOUNDARY = 0.

$\Rightarrow$ STANDING WAVE DEMO DESMOS WAVE AT BOUNDARY
 $c_2 \rightarrow 0.001$



- IF I ADD ANOTHER CLAMP TO A NODE, THE SOLUTION REMAINS VALID, E.G.

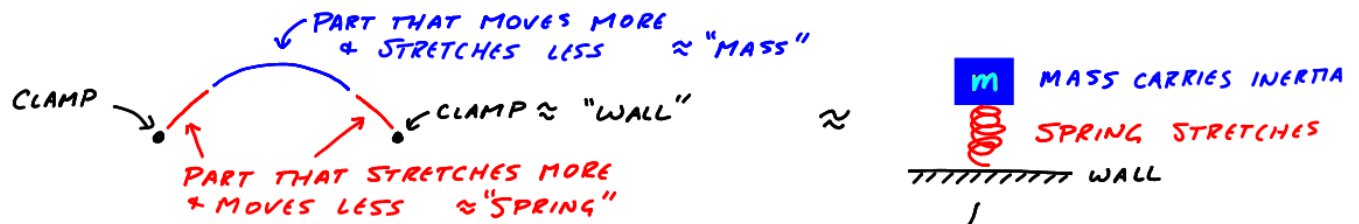


- IF I TRY TO CLAMP ANYWHERE ELSE, THE BOUNDARY CONDITIONS ARE NOT MET (STRING BREAKS!)



$\Rightarrow$ FREQUENCY IS NOT "ALLOWED" BY THE STRUCTURE.

- NOTICE: THE (CORRECTLY) CLAMPED STRING LOOKS LIKE A MASS ON A SPRING!

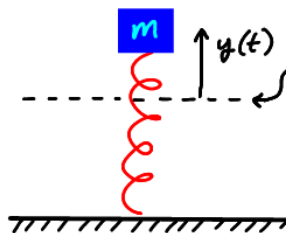


THIS "TOY" SYSTEM WEIRDLY WELL DESCRIBES RESONANCES IN ALL INSTRUMENTS. (INCLUDING SCIENTIFIC!)
- SO LET'S FOCUS ON IT!

HARMONIC OSCILLATOR

MASS + SPRING (AND NO GRAVITY)

WITHOUT DISSIPATION



SIMPLEST MODEL (NO DISSIPATION)

$$F = -Ky = ma \Rightarrow \partial_t^2 y = -\left(\frac{K}{m}\right)y$$

(ENTHUSIASTS)

VERTICAL FORCE SPRING CONSTANT [N/m]

SOLUTION:
 $\Rightarrow y(t) = y_0 \cos\left(\sqrt{\frac{K}{m}}t + \phi_0\right)$ THIS THING JUST OSCILLATES UP AND DOWN INDEFINITELY

CONSTANTS

10pt
Q IF $\sqrt{\frac{K}{m}}$ IS BIG, THIS OSCILLATES _____.

20pt
Q WHAT IS THE TOTAL ENERGY WHEN $y = y_0$?

10pt
Q WHAT IF $y = -y_0$?

20pt
Q WHAT IS THE TOTAL ENERGY WHEN $y = 0$?

END OF WED
OCT 28

10pt
Q WHAT IS THE TIME-DEPENDENT POTENTIAL ENERGY?

20pt
Q WHAT IS THE TIME-DEPENDENT KINETIC ENERGY?

20pt
Q SO WHAT MUST THE TIME-DEPENDENT VELOCITY BE?

WITH DISSIPATION

20pt
Q WITH DISSIPATION, GUESS THE FORM OF $y(t)$?

ENTHUSIASTS / PROFESSIONALS
- ALTERNATIVE FORMS

$$y = y_0 e^{-t/\tau} \cos(\dots)$$

RINGDOWN TIME FOR AMPLITUDE

$$y = y_0 2^{-t/2T_{1/2}} \cos(\dots)$$

HALF LIFE FOR ENERGY

- NOTE HERE WE START THE OSCILLATOR & LET IT RING DOWN WITH TIME.
- FOR $\mathcal{L}$, WE HAD CONTINUOUS DRIVE AT ONE LOCATION AND WATCHED THE WAVE DECAY WITH DISTANCE TRAVELED
- THIS $\mathcal{D}$ IS STRICTLY TIME-DOMAIN DECAY OF OSCILLATIONS AT ONE LOCATION

DEMO DESMOS PLOT 6.1 RINGDOWN $y + v$
- GUESS & WIGGLE PARAMETERS
- CONVERT $\sin \leftrightarrow \cos$ ω / ϕ_0

INITIAL CONDITIONS

- STRING INSTRUMENTS ARE PLUCKED OR STRUCK

PLUCKING: $v = 0, y = y_0$ WHEN RELEASED

^{20pt}
Q WHAT IS ϕ_0 IF WE PLUCK AT $t=0$?

$$y(t) = y_0 \cos(2\pi f_0 t + \phi_0) 10^{-\sigma t/20}$$

$$v(t) = -v_0 \sin(2\pi f_0 t + \phi_0) 10^{-\sigma t/20}$$

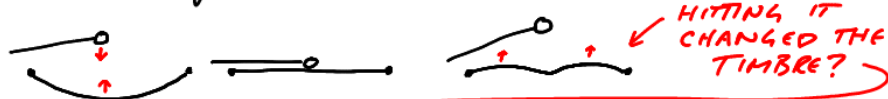
STRIKING: (FROM REST) $v = v_0, y = 0$ IMMEDIATELY AFTER THE STRIKE

^{20pt}
Q WHAT IS ϕ_0 IF STRUCK AT $t=0$?

STRIKING WHILE IT'S MOVING: THE IMPULSE FROM A STRIKE CAN SPEED OR SLOW THE EXISTING MOTION.

- ADJUST $\phi_0 + y_0$ (OR v_0) UNTIL THE INITIAL CONDITIONS ARE MET.

- WE WILL MOSTLY STICK TO SIMPLE CASES, BUT YOU CAN IMAGINE HITTING A DRUM JUST RIGHT + ROUGHLY STOPPING ITS (FUNDAMENTAL) MOTION, e.g.



→ -MAYBE SOME FUN IN HERE LATER ☺

- FUN TO BE HAD WITH GUITARS, TOO ☺

ANOTHER IMPORTANT FIGURE IN MUSIC:

Q FACTOR $\equiv$ " $2\pi \times \text{TOTAL ENERGY} / \text{ENERGY LOST PER CYCLE}$ " $\leftarrow$ WEIRD OBJECT!

- How does "Q" relate to β ?

"NUMBER OF CYCLES BEFORE ENERGY IS MOSTLY GONE"
(EXPONENTIAL DECAY)

20pt

Q

WHAT IS THE TOTAL ENERGY FOR

$$y(t) = y_0 10^{-\beta t/20} \cos(2\pi f_0 t)$$

$$v(t) = -2\pi f_0 y_0 10^{-\beta t/20} \sin(2\pi f_0 t)?$$

- OK, BUT IF YOU PLOT THIS, YOU WILL SEE



NOTICE $m 4\pi f_0^2 = m 4\pi^2 \left(\frac{1}{2\pi} \sqrt{\frac{k}{m}}\right)^2 = k$

$$\Rightarrow E_{\text{tot}} = \frac{1}{2} k y_0^2 10^{-\beta t/10} (\cos^2(2\pi f_0 t) + \sin^2(2\pi f_0 t))$$

$$= \frac{1}{2} k y_0^2 10^{-\beta t/10}$$

INITIAL POTENTIAL
ENERGY ($v(0)=0$)

$$\equiv E_0 10^{-\beta t/10}$$

JUST A SMOOTH DECAY
AS ENERGY SLOSHES
FROM POTENTIAL TO
KINETIC + BACK!

20pt

Q

WHAT FRACTION OF THE ORIGINAL ENERGY IS LEFT AFTER ONE OSCILLATION?

- SO IT LOST $\Delta E_{\text{tot}} = E_{\text{tot}}(T) - E_{\text{tot}}(0)$

$$= E_0 (1 - 10^{-\beta/10 f_0})$$

DEMO PLOT $E = 1 - 10^{-\beta t/10}$ + ZOOM IN TO
SHOW THIS IS LINEAR
FOR SMALL t

- WITH "MATH" (TAYLOR EXPANSION) IT TURNS OUT

$$1 - 10^{-\beta t/10} \approx \left(\frac{\beta}{10} \ln 10 \right) \times t \approx 0.23 \beta t \text{ FOR SMALL } t$$

NATURAL
LOG OF 10
 ≈ 2.3

- SO IN ONE CYCLE,

$$\frac{\Delta E_{\text{TOT}}}{E_{\text{TOT}}} \approx 0.23 \frac{\sigma}{f}$$

⇒ THE "Q FACTOR" IS

$$Q \equiv \frac{2\pi E_{\text{TOT}}}{\Delta E_{\text{TOT}}} = \left(\frac{20\pi}{\ln 10}\right) \frac{f_0}{\sigma} \approx (27.3 \text{ dB}) \frac{f_0}{\sigma}$$

[UNITLESS]
"Q-FACTOR" IN TERMS OF LOSS RATE σ

IMPORTANT: INCREASES w/ f_0
DECREASES w/ σ

- ANOTHER COMMON WAY PEOPLE WRITE THIS

- DECAY FACTOR CAN BE WRITTEN IN TERMS OF "e" INSTEAD OF 10

$$10^{-\sigma t/20} = e^{\ln(10^{-\sigma t/20})} = e^{-\frac{\ln 10}{20} \sigma t} \equiv e^{-t/\tau}$$

≈ 2.7
 $\equiv \frac{1}{\tau}$

DEFINING "TIME CONSTANT"

$$\tau \equiv \frac{20}{\sigma \ln 10}$$

← "HOW LONG IT TAKES FOR AMPLITUDE TO DECAY BY A FACTOR OF e"

NOT NEEDED FOR QUIZZES. BUT YOU MAY SEE THIS IN LIFE!

$$\Rightarrow Q = \pi f_0 \tau$$

"TAU" = "TIME CONSTANT"

- ANOTHER INTERPRETATION OF Q: THE NUMBER OF OSCILLATIONS UNTIL THE ENERGY HAS DECAYED BY SOME FACTOR.

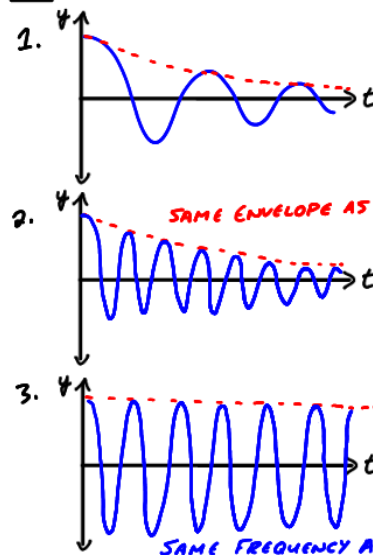
20pt

Q AFTER Q OSCILLATIONS WHAT FRACTION OF THE ORIGINAL ENERGY REMAINS?

END OF WED
OCT 30

20pt

Q HERE ARE 3 OSCILLATORS RINGING:



$Q_1 - Q_2 - Q_3$
 $\sigma_1 - \sigma_2 - \sigma_3$
 $\tau_1 - \tau_2 - \tau_3$

FILL IN THE EQUALITIES/INEQUALITIES

SO FAR WE'VE JUST LET THIS THING RING...
WHAT IF WE CONTINUOUSLY DRIVE IT?

SUSCEPTIBILITY: HOW SUSCEPTIBLE IS A MASS-SPRING SYSTEM TO A FORCE AT FREQUENCY f ?

- THE EQUATION OF MOTION BECOMES

$$m\ddot{y} = \underbrace{-ky}_{\text{NO DISSIPATION, UNDRIVEN}} - \underbrace{\delta \dot{y}}_{\text{"VELOCITY DAMPING"}} + \underbrace{F_0 \cos(2\pi f t)}_{\text{DRIVING FORCE}}$$

CONSTANTS ANY FREQUENCY, NOT THE "NATURAL" $f_0 = \sqrt{\frac{k}{m}}$

"VELOCITY DAMPING"

FORCE THAT REDUCES SPEED / DISSIPATES KINETIC ENERGY
 δ HAS UNITS $[N\cdot s/m] = [kg/s]$

10pt

Q

GUESS WHAT WILL HAPPEN STARTING FROM REST?

"STEADY STATE"
 AMPLITUDE WHEN $t \gg \tau, \frac{1}{\delta}$
 - DISSIPATION BALANCES DRIVE (δ BALANCES F_0)
 - RESPONSE ONLY AT DRIVE FREQUENCY
 - FIGURING THIS "SIMPLE" BEHAVIOR OUT TELLS US A LOT!

- WE CARE ABOUT STEADY STATE ONLY FOR NOW

ENTHUSIASTS: TEST $y(t) = y_0 e^{-i2\pi f t}$ AND $F(t) = F_0 e^{i2\pi f t}$

$$m \partial_t^2 y + \delta \partial_t y + ky = F_0 e^{i2\pi f t}$$

$$\Rightarrow -m(2\pi f)^2 y_0 + i\delta(2\pi f) y_0 + ky_0 = F_0$$

$$\Rightarrow y_0 = \frac{F_0/4\pi^2 m}{f_0^2 - f^2 + i\left(\frac{\delta}{2\pi m}\right)f}$$

MAGNITUDE + PHASE OF THIS IS THE MAGNITUDE + PHASE OF THE RESPONSE!! BECAUSE COMPLEX ANALYSIS RULES.

- CAN ALSO DO THIS ALL WITH $\sin()$ + $\cos()$... THIS IS JUST A SHORTCUT

$\Rightarrow$ THE RESPONSE AMPLITUDE IS

$$y_0(f) = \frac{F_0/4\pi^2 m}{\sqrt{(f_0^2 - f^2)^2 + G^2 f^2}}$$

STEADY STATE AMPLITUDE FOR PURE TONE DRIVE = "SUSCEPTIBILITY"

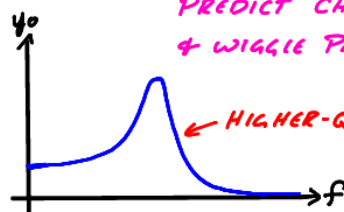
$$G \equiv \frac{\delta}{2\pi m} = \frac{\ln 10}{20\pi} \delta \approx 0.037 \delta$$

DAMPING RATE $[\frac{1}{s}]$

BONUS

DESMOS PLOT

6.2 SUSCEPTIBILITY - DISCUSS $f=0$ + $f=f_0$ + $f>f_0$ REGIMES
 PREDICT CHANGES IN PEAK POSITION, WIDTH, HEIGHT
 + WIGGLE PARAMETERS



HIGHER-Q / LOWER-LOSS OSCILLATORS HAVE NARROWER, TALLER PEAKS!

20pt

Q

MATCHING GAME:

HIGH Q

LONG RINGDOWN

HIGH Q

NARROW RESONANCE

TALL RESONANCE

WIDE RESONANCE

SHORT RESONANCE

WITH FINGERS

DEMO: DAMPED & UNDAMPED TUNING FORK, RECORDED AND CONVERTED TO POWER SPECTRUM. (PYTHON)
OR GUITAR SWEEP W/ HIGH & LOW-LOSS SLIDES.

↑
CHALK

↑
GLASS

- FUN FACTS:

- THIS IS JUST ANOTHER EXAMPLE OF A FREQUENCY-DEPENDENT "FILTER" WE HAVE OCCASIONALLY PLAYED WITH (ABSORPTION, DIFFRACTION, E.G.)
- ⇒ ONCE WE KNOW THE FREQUENCY-DEPENDENT AMPLITUDE $y_0(f)$ AND PHASE $\phi_0(f)$ (NOT DISCUSSED HERE), WE CAN QUICKLY CALCULATE THE RESPONSE TO A MORE COMPLEX DRIVE.
 1. WRITE DRIVE AS SUM OF PURE TONES
 2. APPLY SUSCEPTIBILITY (FILTER)
 3. RE-SUM PURE TONES.

BECAUSE MATH = PRETTY ⇒ THE FOURIER DECOMPOSITION OF FREE EVOLUTION (I.E. RINGDOWN) IS THE SUSCEPTIBILITY

⇒ YOU CAN MEASURE FILTER SUSCEPTIBILITIES BY "KICKING" THEM WITH AN IMPULSE & F.F.T. THE "IMPULSE RESPONSE"

- YOU CAN MODEL FILTERS IN REAL TIME BY FIRST MEASURING THE "IMPULSE RESPONSE" & APPLYING THIS TO EACH INCOMING VALUE:



BONUS: DO THIS

- THIS IS THE "FINITE IMPULSE RESPONSE" (FIR) METHOD.

- RECALL ALL OF THIS IS TO MODEL HOW A SINGLE RESONANCE BEHAVES

BACK TO STRINGS

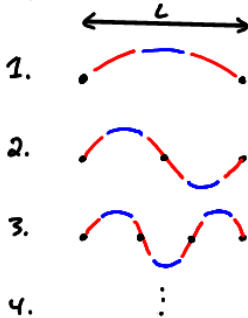
NOW WITH MORE RESONANCES!

RECALL



- FOR THIS, WE NEEDED TO CHOOSE A POINT ON THE STRING (USUALLY THE SPOT OF MAXIMUM DISPLACEMENT) AND WATCH HOW IT OSCILLATES/RESPONDS TO FORCES (AT THAT POINT)

- IN YOUR HOMEWORK YOU WILL FIND MANY RESONANCES HAVING DIFFERENT FREQUENCIES AND WAVELENGTHS



- NOTICE TO HAVE CLAMPED BOUNDARIES WE NEED NODES AT THE ENDS.

⇒ ALLOWED WAVELENGTHS DETERMINED BY FITTING HALF-WAVELENGTHS INTO LENGTH L :

$$L = n \frac{\lambda_n}{2}, \quad n = 1, 2, 3, 4, \dots$$

"nth RESONANCE"

⇒

$$\lambda_n = \frac{2L}{n}$$

⇒

$$f_n = \frac{c}{\lambda_n} = \frac{c}{2L} n$$

CLAMPED STRING RESONANT WAVELENGTHS AND FREQUENCIES

FUN FACT: $\frac{2L}{c}$ IS THE ROUND-TRIP TIME FOR WAVES BOUNCING BACK AND FORTH! ☺

10pt



RESONANT SPATIAL FREQUENCIES?

END OF MON
NOV 4

10pt



DOES THIS LOOK FAMILIAR?

- JUST LIKE HOW AT FIXED x WE CAN WRITE ANY

SAME MATH! DIFFERENT x -AXIS!

$$y(t) = A_1 \cos(2\pi f_1 t + \phi_1) + A_2 \cos(2\pi f_2 t + \phi_2) + \dots \quad (\text{IGNORING } x)$$

AT FIXED TIME, WE CAN WRITE ANY

$$y(x) = A_1 \cos(2\pi \chi_1 x + \phi_1) + A_2 \cos(2\pi \chi_2 x + \phi_2) + \dots \quad (\text{IGNORING } t)$$

10pt



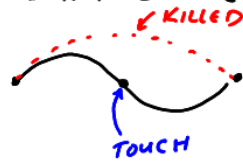
IF $L \rightarrow \frac{L}{2}$, WHAT HAPPENS TO FREQUENCY?

DEMO

SLIDE GUITAR WITH TUNING FORK

- SHOW OCTAVE BY HALVING THE STRING
- TOUCH RINGING FORK TO BASE OF STRING + SLIDE THROUGH RESONANCES
- SLIDE THROUGH 2 NODES + ASK WHAT HAPPENED
- SHOW HARMONIC TUNING.

- BY TOUCHING THE STRING AT NODES OF HARMONICS YOU CAN DAMP OUT EVERY MODE THAT DOESN'T HAVE A NODE THERE.



- NOTE YOU ALSO NEED THE DETECTOR TO NOT SIT AT A NODE ☹

GUITAR: SELECT NECK PICKUP, SHOW LACK OF RESPONSE COMPARED TO OTHERS WITH FINGER ON STRING ABOVE 5th FRET

INCLUDING t & x

- EACH RESONANCE HAS A FORM LIKE

$$y_n(x,t) = \underbrace{A_n}_{\text{HEIGHT}} \underbrace{\cos(2\pi k_n(x-x_n))}_{\text{SPATIAL PROFILE}} \underbrace{\cos(2\pi f_n(t-t_n))}_{\text{UP + DOWN MOTION}}$$

$\swarrow$ SHIFTS NODES $\swarrow$ "START TIME"

- IN THIS CLASS, FOR SIMPLICITY, WE WILL STICK TO THE SITUATION IN WHICH ONE NODE IS AT $x=0$, SO THAT

$$\Rightarrow y_n(x,t) = A_n \sin\left(2\pi k_n x\right) \cos\left(2\pi f_n(t-t_n)\right)$$

$\uparrow$ $\frac{n}{2L}$ $\uparrow$ $\frac{nc}{2L}$

"NORMAL MODES" OF CLAMPED STRING SPANNING $0 \leq x \leq L$
 THIS IS A COMPLETE SET THAT CAN DESCRIBE ALL FREELY MOVING STRINGS!

20pt



IF I PULL THE STRING & RELEASE WHAT WILL HAPPEN?



- THESE HARMONICS HAVE A DEFINITE RELATIONSHIP BETWEEN PROFILE & FREQUENCY

$\Rightarrow$ STRING CANNOT WIGGLE ARBITRARILY!

- WHAT IF WE PLUCK OFF CENTER?

10pt



- SHOW SECOND SUPERPOSITION & ANIMATE ALONE
 IF I PLUCK AT DIFFERENT LOCATIONS, WILL IT AFFECT THE TIMBRE?

END OF
 WED NOV 6

10pt



IF I PUT A DETECTOR (PICKUP OR MIC) CLOSE TO THE STRING WILL THE TIMBRE DEPEND ON POSITION?

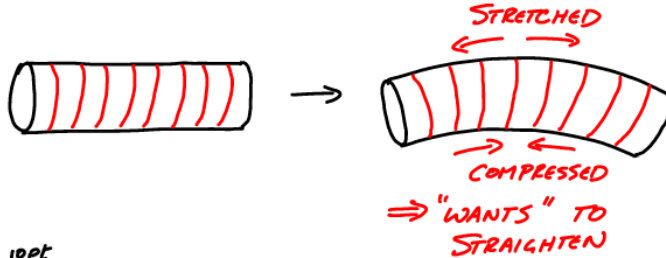
- YOU MAY FIND STRUMMING IN THE MIDDLE MAKES A MORE "CLARINET-LIKE" SOUND WHILE THE EDGE IS MORE "OBOE-LIKE" ☹️ } MORE LATER!

REAL STRINGS

- ON A GUITAR, PIANO, VIOLIN, ETC, STRINGS ARE NOT 1D OBJECTS
- THEY HAVE THEIR OWN STIFFNESS AGAINST BENDING, EVEN WITHOUT ANY TENSION!

DEMO SHOW ENDS OF GUITAR STRINGS

- MORE ACCURATE STRING:



10pt
Q WHAT SHOULD THIS DO TO RESONANCE FREQUENCIES?

- INCREASE
- DECREASE
- NOTHING
- SHUT UP, SANKEY

20pt
Q WILL THIS AFFECT SHORTER OR LONGER WAVELENGTHS MORE?

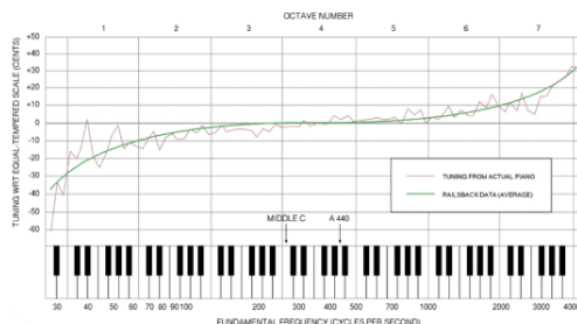
⇒ THE FREQUENCIES OF HIGHER n , SHORTER WAVELENGTH MODES WILL BE A BIT HIGHER THAN THE ACTUAL (MATHEMATICAL) HARMONIC!

- ↪ TUNED WITH PERFECT FREQUENCIES FOR EACH STRING
- ON A PIANO, THE HIGHER n MODES (ACTIVE WHEN STRUCK) FROM THE LOW STRINGS ARE DETUNED RELATIVE TO THE "CORRECT" PITCH OF THE HIGH STRINGS

⇒ SOUNDS AWFUL !!

BEATING + "WARBLING" WHEN PLAYING BOTH EVEN IF THE TUNING IS PERFECT!

SOLUTION:



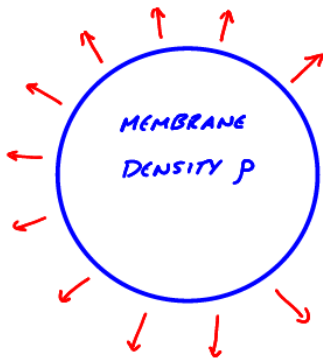
WHERE WE TUNE IT SO IT DOESN'T SOUND LIKE ASS.

← WHERE WE EXPECT THE FREQUENCY

VIOLINS SHARP TO PL

DRUMS

INSTEAD OF A STRING UNDER TENSION (1D), CONSIDER A MEMBRANE UNDER TENSION (2D). *VERY SIMILAR TO A STRING, JUST IN 2D INSTEAD OF 1D*



- ASSUME UNIFORM TENSION FOR SIMPLICITY

- SINCE TENSION IS APPLIED ALONG AN EDGE (RATHER THAN A POINT, AS WITH STRINGS)

TENSION HAS UNITS [N/m]

A WIDER EDGE CAN APPLY MORE FORCE

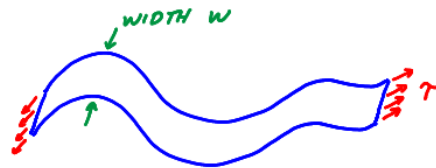
10pt
Q

WHAT UNITS DOES ρ HAVE?

20pt

Q GUESS THE SPEED OF WAVES (BY UNITS). (HINT: RECALL STRING WAVE SPEED)

- RIGOR SHOWS THIS TO BE TRUE, BUT YOU CAN SEE WHY BY IMAGINING A STRIP OF MEMBRANE



- VIEW THIS EDGE-ON AND IT JUST LOOKS LIKE A STRING AGAIN, WITH LINEAR DENSITY

DENSITY PER AREA

$$\rho_{\text{STRIP}} = \rho W \left[\frac{\text{kg}}{\text{m}} \right]$$

AND TENSION

WIDTH: WIDER STRIPS WEIGH MORE

$$F_T = \gamma W \text{ [N]}$$

$\Rightarrow$ WAVE SPEED

$$c = \sqrt{\frac{F_T}{\rho_{\text{STRIP}}}} = \sqrt{\frac{\gamma W}{\rho W}} = \sqrt{\frac{\gamma}{\rho}} \quad \text{!}$$

- AND THE WAVE PHYSICS IS THE SAME AS BEFORE
ALL DIRECTIONS: CURVATURE $\Rightarrow$ ACCELERATION

ENTHUSIASTS: EQUATIONS OF MOTION

$$\text{STRING: } \rho \partial_t^2 y = F_T \partial_x^2 y$$

$$\text{MEMBRANE: } \rho \partial_t^2 y = \underbrace{\gamma \partial_x^2 y}_{\text{"ma"}} + \underbrace{\gamma \partial_z^2 y}_{\text{"F"}}$$

CURVATURE ALONG THE OTHER IN-PLANE DIRECTION (z) ALSO CONTRIBUTES!

TRAVELING WAVES

- THE NATURAL SOLUTIONS ARE THEN PLANE WAVES IN 2D:

VERY SIMILAR
TO WAVES
ON A STRING!

$$y(x, z, t) = y_0 \cos(2\pi(\underbrace{k_x x + k_z z}_{\text{NOTICE A DOT PRODUCT? :D}} - ft) + \phi_0)$$

INFINITE
MEMBRANE TRAVELING
WAVE SOLUTIONS

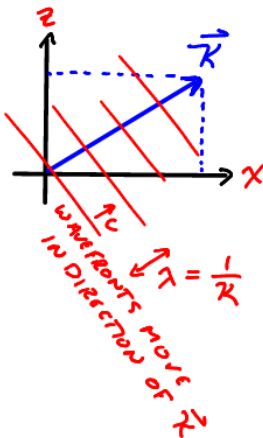
- IF WE TRY THIS SOLUTION, WE FIND

$$-pf^2 = (-k_x^2 - k_z^2)\gamma \Rightarrow \frac{\gamma}{\rho} = \frac{f^2}{k_x^2 + k_z^2}$$

$$\Rightarrow c = \frac{f}{k} = \frac{f}{\sqrt{k_x^2 + k_z^2}}$$

WHERE k IS THE MAGNITUDE
OF THE VECTOR $\vec{k} = (k_x, k_z)$

WAVE SPEED, FREQUENCY,
AND SPATIAL FREQUENCY
FOR A 2D WAVE.



VERY COOL! $\vec{k}$ IS
THE DIRECTION ALONG
WHICH THE WAVEFRONTS
TRAVEL WITH SPEED c
AND SPATIAL FREQUENCY $k = |\vec{k}|$
(AND WAVELENGTH $\lambda = \frac{1}{k}$)

- THANK GOODNESS, TOO, BECAUSE
WE COULD ALWAYS DEFINE OR
"RIBBON AXIS" ALONG $\vec{k}$ + IT
BETTER BEHAVE LIKE A STRING!

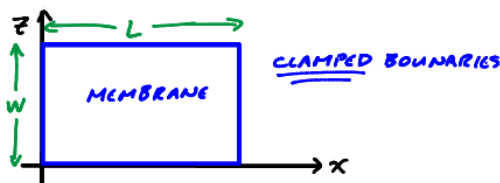
STANDING WAVES

- SIMILAR TO THE STRING, WE CAN GET STANDING WAVES BY COMBINING THESE SOLUTIONS.
 $\Rightarrow$ NODES, ANTINODES, ETC... $\Rightarrow$ MATH + TRIG IDENTITIES IN 2D

- OR WE CAN JUST GUESS THE SOLUTIONS FROM OUR KNOWLEDGE OF STRING BEHAVIOR.

4066

Q FOR THE RECTANGULAR MEMBRANE BELOW, GUESS THE SOLUTIONS



- THESE ARE CORRECT SOLUTIONS! (TRY IN DIFF EQ'S) (TRY PLOTTING!)

- STANDING WAVE IN $x \Rightarrow$ NODES ALONG z

- STANDING WAVE IN $z \Rightarrow$ NODES ALONG x

- $\vec{k}$ VECTOR FOR WAVE "BOUNCING AROUND A BOX" (RATHER THAN "BETWEEN TWO POINTS" ON A STRING)



20pt



ARE THE RESONANT FREQUENCIES HARMONICS OF THE FUNDAMENTAL?

$\Rightarrow$ YOU WILL GET SOME CRAZY NON-REPETITIVE OSCILLATIONS AND INTERESTING TIMBRE. :)

$\Rightarrow$ WHY THEY AREN'T USED FOR MELODY?

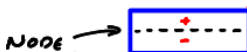
END OF
MON NOV 11

20pt



MATCHING GAME, AND ORDER THE FREQUENCIES LOWEST TO HIGHEST (1 TO 4)

MODE



$$n_x = 1$$

$$n_z = 1$$

$$n_x = 2$$

$$n_z = 2$$

INTUITION: ADDING NODES IN TIGHTLY CONFINED DIRECTIONS REQUIRES BIG FREQUENCY CHANGES

(+ = RELATIVE DIRECTION UP OR DOWN)

OTHER DRUMS: SAME GAME, DIFFERENT COORDINATES.

- DIFFERENT SHAPES HAVE DIFFERENT RESONANCES, AS DEFINED BY

- SHAPE OF THE BOUNDARY

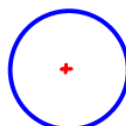
- SPEED OF WAVES (TENSION, DENSITY)

PLUS FURTHER FILTERING + (LOW Q!) RESONANCES FROM THE 3D STRUCTURE + CHAMBER THEY ARE ATTACHED TO.

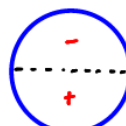
- USUALLY THEY ARE ROUND.

- INSTEAD OF ADDING NODES ALONG $x+z$, WE DRAW THEM ALONG RADIUS (r) AND ANGLE (θ):

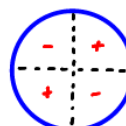
BONUS: SIMULATE A TIMPANY!



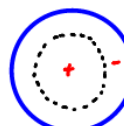
$$f_0$$



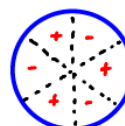
$$1.59f_0$$



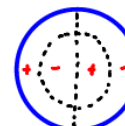
$$2.14f_0$$



$$2.30f_0$$



$$2.65f_0$$



$$2.92f_0$$

$$y(r, \theta) = y_0 R_{ln}(r) \cos(l\theta)$$

SPATIAL PART

"BESSEL FUNCTIONS" (KINDA LIKE COSINE)

THE THING YOU'RE USED TO, JUST ROUND!

20pt



$$l=0, n=0$$

$$l=, n=$$

$$l=, n=$$

$$l=, n=$$

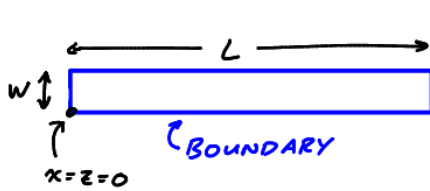
$$l=, n=$$

$$l=, n=$$

DEMO 6.4 MEMBRANE
3D + EDGE ON

RESONANT CHAMBERS & PIPES

CONSIDER A NARROW PIPE WITH CLOSED ENDS



20pt
Q WHAT ARE THE BOUNDARY CONDITIONS FOR SOUND WAVES AT THE WALLS?

ΔP CLAMPED (NODE)
 u FREE (ANTINODE)

20pt
Q WHAT WAVELENGTHS "FIT" ALONG THE LENGTH OF THE PIPE? (IGNORE THE OTHER DIRECTION FOR NOW)

20pt
Q ANY OTHER SOLUTIONS?

THIS IS GREAT, BECAUSE IT ALLOWS FOR SOLUTIONS OF THE FORM

CONSTANT ALONG y & z .
THINK $\Delta P_0 \cos(0 \cdot y) \cos(0 \cdot z)$

$$\Delta P = \Delta P_0 \cos(2\pi k_n x) \cos(2\pi f_n t + \phi_n)$$

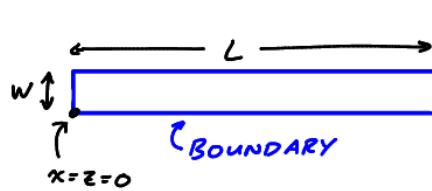
THIS CONSTANT IS JUST THE $n=0$ SOLUTIONS FOR THE OTHER DIMENSION(S)

$\Rightarrow$ THIS CAN BECOME A $\sim 1D$ PROBLEM!!
(SO LONG AS $\lambda > 2W$ SO WE CANNOT FIT THE WAVE IN THE OTHER DIMENSION)

20pt
Q WHAT VALUES OF k MAKE THIS STANDING WAVE?

IMPORTANT:

- NOT POSSIBLE WITH CLAMPED MEMBRANE ($\sin(0 \cdot x) = 0$)
- MAKES HARMONIC RESONANCES POSSIBLE DESPITE BEING A 3D CHAMBER !!



20pt
Q

WHAT ARE THE BOUNDARY CONDITIONS FOR SOUND WAVES AT AN OPEN END?

ΔP
 w

NODE
ANTINODE

20pt
Q

WHAT ARE THE RESONANT FREQUENCIES NOW?

20pt
Q

ARE THE RESONANT FREQUENCIES HARMONICS?

20pt
Q

WHAT ABOUT AN OPEN-OPEN PIPE?

DEMO

OPEN-OPEN.WAV
OPEN-CLOSED.WAV

} ADDED ALL OVERTONES
UP TO 10 kHz

END OF
WED NOV 13

- WHICH SOUNDS MORE LIKE A CLARINET?

OPEN PIPE ORGAN?

OTHER INSTRUMENTS?

OBOE? (WE WILL COME BACK TO THIS)

- BUT, AS MENTIONED, THE OPEN END IS CLEARLY NOT 100% CLAMPED ΔP , SINCE WE KNOW THESE EMIT PRESSURE WAVES!

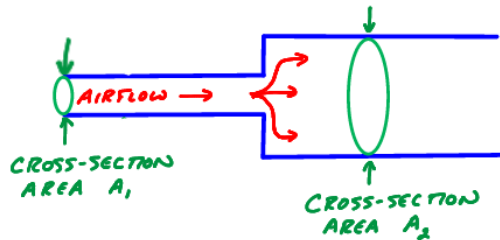
LOOK UP "END CORRECTION", WHICH WE IGNORE.

- LET'S LEARN A BIT MORE BY CONSIDERING LESS EXTREME BOUNDARY CONDITIONS...

AIRFLOW IN TUBES

- FOR SOUND WAVES IN TUBES, IT IS IMPORTANT TO DISCUSS
"AIR FLOW RATE" (VOLUME PER SECOND) [m^3/s]

- CONSIDER A BOUNDARY BETWEEN TWO PIPES



TOY MODEL

- SINCE AIR HAS TO GO SOMEWHERE, THE FLOW RATE MUST BE THE SAME ON BOTH SIDES OF THE BOUNDARY
- THE PRESSURE SHOULD ALSO HAVE THE SAME VALUE ON BOTH SIDES (OR ELSE THE THIN SLABS OF AIR AT THE BOUNDARY WOULD MOVE TO EQUALIZE IT.)

20pt

Q FOR A GIVEN FLOW VELOCITY v AND CROSS-SECTION AREA A , WHAT VOLUME OF AIR PASSES IN TIME Δt ?

- WE DEFINE AIR FLOW RATE AS PASSING VOLUME PER TIME

$$\frac{\Delta V}{\Delta t} = A v \quad \text{AIR FLOW RATE } (\text{m}^3/\text{s})$$

Annotations:
- ΔV is labeled "VOLUME" with a red arrow.
- Δt is labeled "TIME" with a red arrow.
- A is labeled "CROSS-SECTION AREA" with a red arrow.
- v is labeled "VELOCITY" with a red arrow.

20pt

Q IF THE VELOCITY IN TUBE 1 IS v_1 , WHAT IS THE VELOCITY IN TUBE 2?

- BUT THE PRESSURES AT THE BOUNDARY BETWEEN THE PIPES MUST BE CONTINUOUS!

⇒ SOUND WAVES WILL HAVE DIFFERENT VELOCITY AMPLITUDES FOR A GIVEN PRESSURE AMPLITUDE
⇒ DIFFERENT IMPEDANCES?

20pt

Q SHOULD THE SOUND SPEED BE THE SAME IN EACH PIPE?

- AS IT TURNS OUT, THE "IMPEDANCE" FOR A PIPE OF CROSS-SECTION AREA "A" IS

BONUS:
SHOW THIS
WITH CONTINUITY
OF FLOW RATE.

$$Z = \frac{\rho c}{A}$$

CHARACTERISTIC IMPEDANCE OF SOUND PLANE WAVES IN A PIPE OF AREA "A".

USED TO CALCULATE REFLECTED + TRANSMITTED PRESSURE AMPLITUDES.

→ IMPORTANT:

SINCE (UNLIKE FREE AIR) THE VELOCITY AMPLITUDE DEPENDS ON BOTH PRESSURE AMPLITUDE AND AREA, WE CAN ONLY USE THEM TO CALCULATE REFLECTED + TRANSMITTED POWERS (NOT INTENSITY!)

i.e.

$$\begin{aligned} P_{\text{REFL}} &= \left(\frac{Z_2 - Z_1}{Z_2 + Z_1} \right)^2 P_{\text{IN}} \\ P_{\text{TRANS}} &= \frac{4Z_1 Z_2}{(Z_2 + Z_1)^2} P_{\text{IN}} \end{aligned}$$

USE THIS FOR PIPES, THEN WE CAN GET INTENSITY FROM POWER/AREA.

- VERIFIED EASILY THAT $P_{\text{IN}} = P_{\text{REFL}} + P_{\text{TRANS}}$ FOR AIR IN PIPES:

$$\begin{aligned} \Rightarrow \frac{P_{\text{TRANS}} + P_{\text{REFL}}}{P_{\text{IN}}} &= \frac{4Z_1 Z_2}{(Z_2 + Z_1)^2} + \frac{(Z_2 - Z_1)^2}{(Z_2 + Z_1)^2} \\ \text{(SAME } \rho c \text{'S CANCEL!)} &= \frac{4 \frac{1}{A_1 A_2}}{\left(\frac{1}{A_1} + \frac{1}{A_2} \right)^2} + \frac{\left(\frac{1}{A_2} - \frac{1}{A_1} \right)^2}{\left(\frac{1}{A_1} + \frac{1}{A_2} \right)^2} \\ &= \frac{\frac{1}{A_1^2} - 2\frac{1}{A_1 A_2} + \frac{1}{A_2^2} + 4\frac{1}{A_1 A_2}}{\frac{1}{A_1^2} + \frac{1}{A_2^2} + 2\frac{1}{A_1 A_2}} = 1 \end{aligned}$$

NOTES:

- THIS IS THE SAME AS FOR "FREE" WAVES BUT WITH FREE WAVES THE "AREA" IS THE SAME ON EITHER SIDE OF THE BOUNDARY, (MAKING POWER VS INTENSITY MODT)

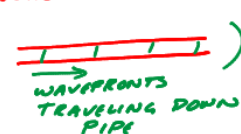
OR WE CONSIDER A CHUNK OF AREA AT THE BOUNDARY.

- WE HAVE USED THE LOWEST-ORDER RESONANCE IN THE TRANSVERSE DIRECTIONS BUT THERE EXIST WAVES WITH NODES RUNNING ALONG THE PIPE (ABOVE FREQUENCIES WHERE λ IS COMPARABLE TO THE LATERAL DIMENSIONS)

- LIMITATIONS OF THIS PICTURE

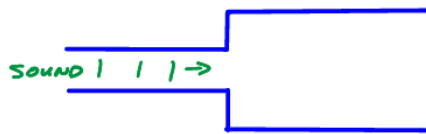
- AIR IS COMPRESSIBLE + THERMODYNAMIC
- WHEN FLOWING INTO A NEW PIPE, THERE IS A REGION OF TURMOIL BEFORE STEADY FLOW IS REACHED
- THE WALLS APPLY DRAG

WE COULD VIEW THE END OF A HORN AS A BOUNDARY TO AN $A \rightarrow \infty$ PIPE, AND ΔP IS NOT QUITE ZERO AT THE EXIT!



ΔP FALLS OFF "QUICKLY" AFTER THE EXIT
⇒ PIPE "SEEMS" A LITTLE LONGER

INTUITIONS

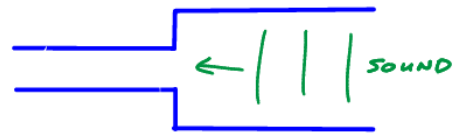


20pt

Q

IF WE SEND A SOUND WAVE IN AS SHOWN,

REFLECTED PRESSURE AMPLITUDE	SAME SIGN
REFLECTED VELOCITY AMPLITUDE	OPPOSITE SIGN
TRANSMITTED PRESSURE AMPLITUDE	ZERO
TRANSMITTED VELOCITY AMPLITUDE	BIGGER
	SMALLER



20pt

Q

IF WE SEND A SOUND WAVE IN AS SHOWN,

REFLECTED PRESSURE AMPLITUDE	SAME SIGN
REFLECTED VELOCITY AMPLITUDE	OPPOSITE SIGN
TRANSMITTED PRESSURE AMPLITUDE	ZERO
TRANSMITTED VELOCITY AMPLITUDE	BIGGER
	SMALLER

- THAT'S IT! NOW GO MAKE SOME STANDING WAVES!
 DESIGN SOME PIPES! (ALMOST... SEE CONE-SHAPED PIPES
 IN 7. FEEDBACK)

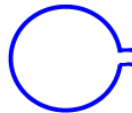
HELMHOLTZ RESONATOR: BIG CAVITY + SMALL PIPE(S)

- YOU WILL ALSO RUN ACROSS ACOUSTIC RESONATORS WHERE THE ABOVE ANALYSIS FAILS
 BECAUSE THEY OPERATE IN A DIFFERENT LIMIT.

DEMO: LOW LOON CALL WITH HANDS

- FREQUENCY $\sim 400 \text{ Hz} \Rightarrow \lambda \sim 1 \text{ m}$
- EVEN A QUARTER λ IS WAY TOO BIG!
- SO WHAT'S DIFFERENT?

MY HANDS $\approx$



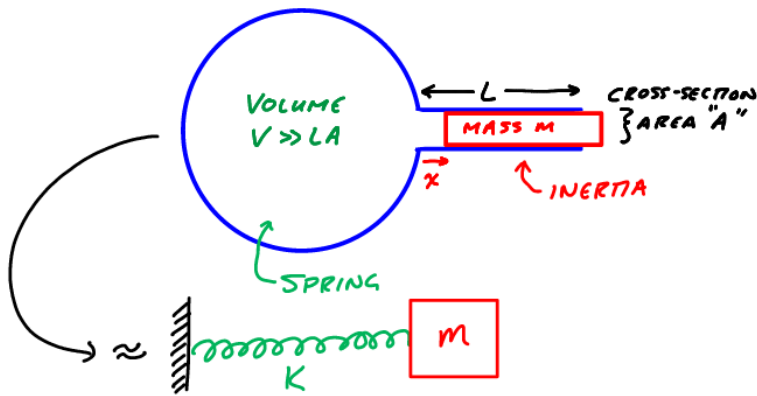
← PITCH CONTROLLED
 BY HOLE SIZE
 CHAMBER SIZE...

- OTHER SUCH RESONATORS:

- OCARINA
- UDU DRUM
- ACOUSTIC GUITAR / VIOLIN BODY
- BOTTLES + JUGS
- DJEMBE BODIES?
- YOUR MOUTH?
- A ROOM + A HALLWAY?

END OF
MON NOV 18

- CONSIDER THIS SHAPE



- MASS M IN NECK FREE TO SLIDE

- VOLUME V OF AIR IN CHAMBER ACTS LIKE SPRING

(COMPARATIVELY LOW VELOCITIES IN THERE ≈ 0 KINETIC ENERGY)

⇒ RESONANT FREQUENCY LIKE USUAL HARMONIC OSCILLATOR

$$f_0 = \frac{1}{2\pi} \sqrt{\frac{K}{M}} \quad \leftarrow \text{SPRING CONSTANT FROM AIR IN CHAMBER}$$

$$= \frac{1}{2\pi} \sqrt{\gamma \frac{A^2 P}{M V}}$$

IF MASS IS AIR, THEN $M = \rho LA$

THERMODYNAMICS $\Rightarrow \gamma \frac{P}{V}$
(SAME AS NOTES 1)
 $= \frac{C_P}{C_V} \approx 1.4$

($\sqrt{1.4} \approx 1.2$)

$$\Rightarrow f_0 = \frac{1}{2\pi} \sqrt{\gamma \frac{P}{\rho} \frac{A}{LV}}$$

$= c^2$:)

$$\Rightarrow f_0 = \frac{c}{2\pi} \sqrt{\frac{A}{LV}} \quad \text{HELMHOLTZ RESONANT FREQUENCY}$$

BONUS: CONNECT THIS TO THE PIPE PICTURE



- LIMITATIONS :

- ASSUMES NECK AIR DOES NOT COMPRESS/EXPAND

- ASSUMES BODY IS SO BIG THAT ITS AIR BASICALLY DOESN'T MOVE COMPARED TO THAT OF THE NECK.

- IGNORES DRAG FROM WALLS

20pt
Q

IF WE ADD ANOTHER IDENTICAL PIPE TO THE BODY, RESONANT FREQUENCY
INCREASES
DECREASES
STAYS THE SAME

DEMO HAND OCARINA, OPEN 2ND FINGER HOLE.

- TRY CUPPING YOUR HANDS OVER YOUR EAR AND LISTEN TO THE PITCH CHANGE WITH GEOMETRY

BUT HOW IS THIS PRODUCING SOUND?

⇒ NEXT NOTES!

SUMMARY

- STANDING WAVE RESONATORS

- IN EACH DIMENSION:

- OPEN BOUNDARY $\Rightarrow$ ANTINODE

- CLAMPED BOUNDARY $\Rightarrow$ NODE

- FIT WAVES INSIDE:

OPEN-OPEN OR CLAMPED-CLAMPED

(n HALF-WAVELENGTHS)

$$K_n = \frac{n}{2L} \quad \text{SPATIAL FREQUENCY}$$

DISTANCE BETWEEN BOUNDARIES

OPEN-CLAMPED (QUARTER WAVELENGTH + n HALF-WAVELENGTHS)

$$K_n = \frac{n + \frac{1}{2}}{2L} \quad \text{IN 1D, } f_n = c K_n$$

- SOLUTIONS OF THE FORM

$$g_n(x, t) = g_{n0} 10^{-\delta t/20} \cos(2\pi K_n x + \phi_n) \cos(2\pi f_n t + \psi_n)$$

g_{n0} : LOSS STARTING AMPLITUDE.
 $10^{-\delta t/20}$: AMPLITUDE $\sim$ (POWER)^{1/2}
 $\cos(2\pi K_n x + \phi_n)$: SPATIAL PROFILE
 $\cos(2\pi f_n t + \psi_n)$: OSCILLATION
 ϕ_n : SETS WHERE NODES ARE
 ψ_n : SETS INITIAL CONDITION
 $g_n(x, t)$: COULD BE PRESSURE, VELOCITY, DISPLACEMENT
 x, y, z : COULD BE x, y, z

- IN MULTIPLE DIMENSIONS (RECTANGULAR BOUNDARIES)

- SOLUTIONS HAVE A PRODUCT OF SPATIAL PROFILES, e.g., IN 3D

$$g_{n_x n_y n_z}(x, y, z, t) = g_{n_x n_y n_z 0} 10^{-\delta t/20} \cos(2\pi K_{n_x} x + \phi_{n_x}) \cos(2\pi K_{n_y} y + \phi_{n_y}) \cos(2\pi K_{n_z} z + \phi_{n_z}) \cos(2\pi f_{n_x n_y n_z} t + \psi_{n_x n_y n_z})$$

$g_{n_x n_y n_z 0}$: STARTING AMPLITUDE
 $10^{-\delta t/20}$: LOSS
 $\cos(2\pi K_{n_x} x + \phi_{n_x}) \cos(2\pi K_{n_y} y + \phi_{n_y}) \cos(2\pi K_{n_z} z + \phi_{n_z})$: SPATIAL PROFILE
 $\cos(2\pi f_{n_x n_y n_z} t + \psi_{n_x n_y n_z})$: OSCILLATION

WITH

$$f_{n_x n_y n_z} = c K_{n_x n_y n_z} = c \sqrt{K_{n_x}^2 + K_{n_y}^2 + K_{n_z}^2}$$

LENGTH OF $\vec{K}_{n_x n_y n_z}$

PICK A POINT IN SPACE AND EACH OF THESE BEHAVES LIKE A HARMONIC OSCILLATOR:

$\approx \frac{m}{2} K$ EFFECTIVE MASS + SPRING
 WITH $f_0 = \frac{1}{2\pi} \sqrt{\frac{K}{m}}$ NATURAL/RESONANT FREQUENCY

$$Q \equiv \frac{2\pi E_{\text{TOT}}}{\Delta E_{\text{TOT}}} = \left(\frac{20\pi}{\ln 10}\right) \frac{f_0}{\delta} \approx (27.3 \text{ dB}) \frac{f_0}{\delta}$$

[UNITLESS]
"Q-FACTOR" IN TERMS OF LOSS RATE δ

$\Rightarrow$ NO MORE NEED FOR SPATIAL PROFILE!

$$= 2\pi \frac{E_{\text{TOTAL}}}{\Delta E_{\text{CYCLE}}}$$

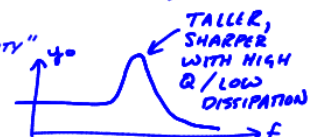
- SUSCEPTIBILITY TO PURE TONE FORCE OF AMPLITUDE F_0 (STEADY STATE)

$$y_0(f) = \frac{F_0 / 4\pi^2 m}{\sqrt{(f_0^2 - f^2)^2 + G^2 f^2}}$$

STEADY STATE AMPLITUDE FOR PURE TONE DRIVE = "SUSCEPTIBILITY"

$$G \equiv \frac{\delta}{2\pi m} = \frac{\ln 10}{20\pi} \delta \approx 0.037 \delta$$

DAMPING RATE $[\frac{1}{s}]$



PIPES:



- HARD WALLS ENABLE SIMPLE "1D-LIKE" SOLUTIONS
- IMPEDANCE

$$Z = \frac{\rho c}{A}$$

FOR CALCULATING REFLECTION & TRANSMISSION BETWEEN PIPES.

} FOR CALCULATING PRESSURE AMPLITUDES AND POWERS ONLY

$$\frac{\Delta V}{\Delta t} = A w$$

AIR FLOW RATE (VOLUME PER SECOND [m^3/s])

HELMHOLTZ RESONATORS: ANOTHER HARMONIC OSCILLATOR WITH CHAMBER FOR SPRING AND NECK AIR FOR MASS.

$$f_0 = \frac{c}{2\pi} \sqrt{\frac{A}{LV}}$$

AREA OF NECK

HELMHOLTZ RESONANT FREQUENCY

VOLUME OF CHAMBER

LENGTH OF NECK

